

SWIFTUI · SCENEKIT · IPHONE DUO HINGE

FOLDSPACE



Fold the phone. Fold space.

BITRIG HACKS · IPHONE DUO EDITION · NYC · SEPT 26 2026

Hinge-angle sensors have existed for years.
Everyone used them for animations.

We made the hinge
an input device.

And the two screens
a 3D corner.

The posture is the state.

No mode switch. No button. Five angle bands, read live from the hinge, with hysteresis so the UI never flickers at a boundary.



135–170° IS OPEN: A WIDER COCKPIT, STILL FLYING

Six gestures. Six verbs.

One recogniser: a 2.5-second ring of angle, velocity and time, with hysteresis, so a squeeze never reads as a warp and a hop never reads as a close.

Close the phone

`.warpClose · 60–160°/s`

WARP

Squeeze, then snap it open

`.squeezeBegan → .snapOpen · > 250°/s`

FIRE

Dip the lid and bring it back

`.hop`

HOP

Pump it four times, then slam

`.pump(count:) → .pumpComplete`

FOLD CORE

Close slowly near the Sun

`continuous angle · depth = fold`

DIVE

Lay it flat

`.laidFlat`

MAP

The planet sits in the crease.

Camera driven by the live hinge angle.

One SceneKit scene, one camera; its pitch is the hinge angle, every frame. Open the phone and the planet stays put while the windshield tilts away. Only the probe may cross the fold.



WINDSHIELD

FOLD · division

CONSOLE

Fold space.

Alcubierre, taken literally: contract space
ahead, expand it behind, ride the bubble in the
crease.

BUBBLE STABILITY = CLOSE SPEED

< 60°/s

60–160°/s

> 160°/s

CRAWL

STABLE

SLAM

OUTER DISPLAY, IN TRANSIT

FOLDING SPACE · 4.24 ly

BEHIND — SPACE EXPANDS

AHEAD — SPACE CONTRACTS

YORK TIME θ , $v \rightarrow +x$

Dive into the Sun.

CORONA ~1,000,000 K

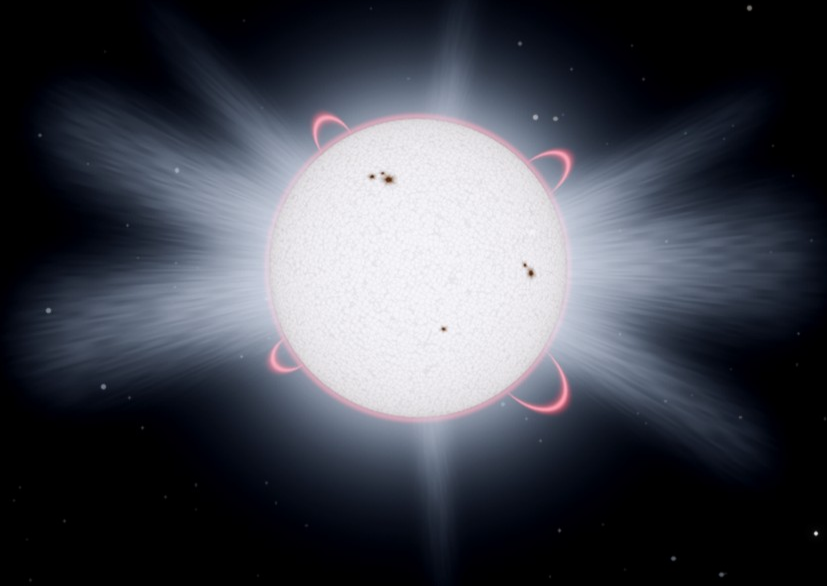
PHOTOSPHERE 5,800 K

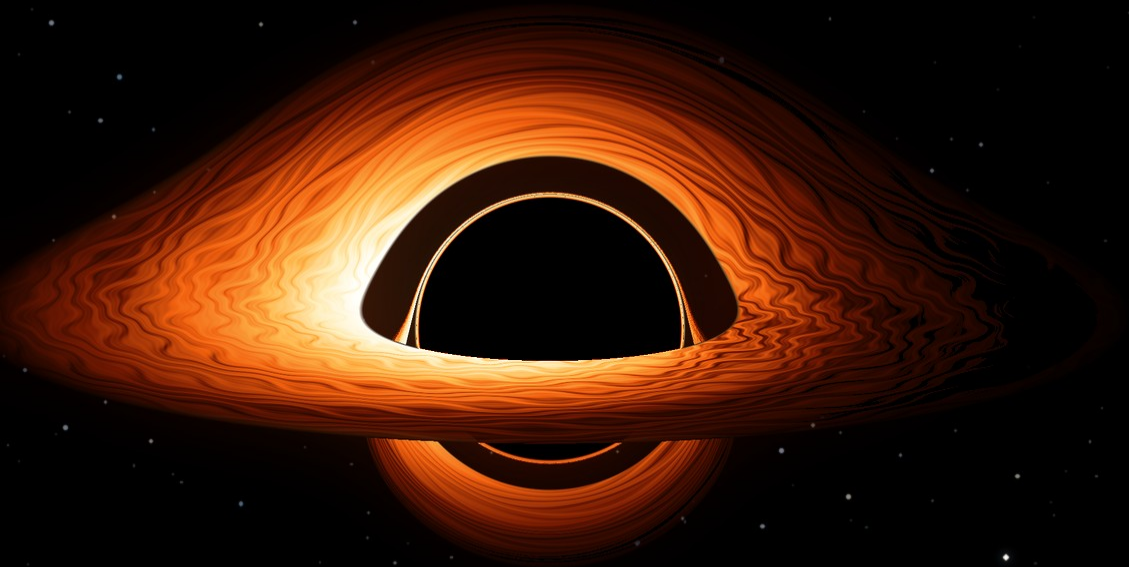
CORE 15,000,000 K

Depth is fold amount. Close slowly and the temperature climbs; open to climb out. The core sample unlocks the Nova Lance.

NOVA LANCE

Squeeze and hold: the crease glows. Snap it open: the planet shatters in the corner.





Sagittarius A^{*}

PHOTON RING

$1.5r_s$

DOPPLER BEAMING

Approaching side brighter

ISCO

$3r_s$

LENSING

Far-side disc over / under the shadow

FOLDSPACE

Andromeda.

2.5 million light-years.

Pump the hinge. Slam it shut.



Built with iPhone Duo APIs.

IPHONE DUO

onHingeChange → DeviceHinge.angle

reservedRegions(.division / .occlusion)

Size classes, split at the fold

Outer ↔ inner display continuity

DUO_SDK flag: Duo hinge, sim, or tilt

ENGINE + SCIENCE

SwiftUI + SceneKit

Real astronomy: NASA Exoplanet
Archive, EHT

26 procedural Blender worlds

Schwarzschild-lensed Sgr A*

SPONSOR STACK

Supabase — Galactic Registry

OpenAI — the ship computer

Sentry — crashes, 60 Hz hinge loop

26 WORLDS



09 — WHY THIS COULDN'T EXIST BEFORE

A FLAT IPHONE HAS

No angle.

Nothing to close, squeeze, snap or pump.

One plane.

No corner for a hologram to sit in.

One face.

Nothing to show FOLDING SPACE to the world.

iPhone Duo has all three, and tells the app about them continuously.

[GITHUB.COM/VNMOORTHY/FOLDSPACE](https://github.com/vnmoorthy/foldspace)

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